

from being fully exploited [26, 36]. Differences in visual appearance and physical dynamics cause policies trained in simulation to degrade when deployed on real hardware.

This gap can be decomposed into two coupled challenges. The first is visual domain shift. Policies that consume synthetic RGB images tend to overfit to rendering-specific textures and lighting, causing them to fail when visual conditions change [13, 36]. A natural remedy is to rely on depth sensors, which are less sensitive to appearance, but commodity depth sensors produce noisy and incomplete readings and often fail on transparent or reflective surfaces [32]. Since geometric properties such as object shape and surface orientation are largely shared between simulation and reality [13], a more promising direction is to extract geometric representations directly from monocular RGB, bypassing both the synthetic texture problem and the physical sensor limitations.

The second challenge is dynamics mismatch. A standard approach is to train a reinforcement learning (RL) expert under automatic domain randomization (ADR) [1], which exposes the policy to a wide distribution of physical parameters. However, distilling this expert into a deployable policy is non-trivial: conventional behavior cloning assumes a unimodal action distribution and averages over the diverse behaviors induced by different randomization parameters, producing suboptimal actions [7]. Moreover, policies that output single-step commands tend to oscillate during contact-rich manipulation [5, 41]. Addressing these two challenges in isolation is insufficient, as visual failures and control instabilities compound each other during real-world execution. A viable solution must handle both perception and control within a unified framework.

We propose **GEARS** (**GE**ometric **A**ction **R**epresentation for **S**im-to-real), a unified framework that jointly addresses the visual and dynamics gaps. For perception, we fine-tune a pretrained visual foundation model [24, 37] to jointly predict monocular depth and surface normals using privileged simulation labels. By supervising both tasks from a shared representation and enforcing a cross-task consistency constraint, we encourage the encoder to internalize the 3D structure of the scene rather than relying on texture cues, producing features that are largely invariant to the visual domain shift. For control, we adopt a teacher-student paradigm [4] in which a privileged RL expert serves as the teacher and a diffusion policy [5] serves as the student. On the teacher side, we propose an efficient particle-based ADR strategy that steers the randomization distribution toward difficult regions, enabling the expert to acquire robust behaviors more sample-efficiently. On the student side, the diffusion formulation models the full action distribution rather than a single mode, naturally accommodating the behavioral diversity introduced by randomization. It also generates temporally coherent action sequences rather than per-step commands, reducing the oscillations common during contact [5]. A central design principle is that these two components are mutually beneficial: the geometric encoder provides a stable, domain-invariant observation space that simplifies the learning problem for the diffusion policy, while the generative policy is expressive enough to exploit the rich spatial information in these representations.